



Finding the hazards a map cannot see

A plain-language guide to HATI:
what it measures, what the plots show, and what the numbers mean

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1 What this document is

This is the guide for anyone who wants to understand HATI without a background in planetary science. It walks through each figure the project produces, says what you are looking at, and explains what every number means and why it matters. No prior knowledge is assumed. Technical terms are defined the first time they appear and again in the glossary at the back.

The short version: HATI looks at pictures and elevation maps of the Moon taken from orbit, and tries to work out which patches of ground would be dangerous to land on. The interesting part is that the most dangerous things are usually too small for the maps to show, so HATI has to find them indirectly.

2 Why this exists

On 6 March 2025 a spacecraft called IM-2, nicknamed Athena, landed near the Moon's south pole and tipped over. Analysis afterwards pointed to a crater about 20 m across on gently sloped ground, right where it set down.

That is the whole problem in one sentence. The elevation maps used to choose the landing site had a resolution of about 4 m per pixel, and in practice they cannot resolve features smaller than roughly four times that. A shallow 20 m crater sits right at the edge of what such a map can show, and a half-metre boulder is completely invisible to it.

So HATI asks a narrow, checkable question. Using only the data that existed before the landing, could an honest, physics-based method have flagged that kind of hazard?

In plain words

An elevation map is like a very coarse relief globe. It shows mountains and big craters fine. It cannot show you a rock the size of a football, and a rock that size is enough to tip a lander. HATI is an attempt to find those small things without pretending the map can see them.

3 The one idea behind HATI

Hazards live at two very different sizes, so HATI uses two separate methods.

Big terrain means slopes and drops a lander physically cannot straddle. That is visible in an elevation map, and measuring it is straightforward geometry.

Small obstacles means rocks and shallow pits below the map's resolution. These are invisible in elevation, but they are visible in a photograph, because they cast *shadows*. Near the Moon's poles the Sun sits only a few degrees above the horizon, so even a small rock throws a long shadow. The geometry does the magnification for you.

The rule that connects a shadow to the rock that cast it is simple trigonometry:

$$L = h \cdot \cot(e) \quad (1)$$

where L is the shadow's length, h is the object's height, and e is the Sun's angle above the horizon. At the pole e is often around 5° , which makes $\cot(e)$ about 11. A rock 0.3 m tall therefore casts a shadow roughly 3.4 m long. The rock is a third of a pixel wide. Its shadow covers several pixels, and several pixels is something you can measure.

One design decision runs through everything. Every HATI result must be traceable to a stated physical quantity that somebody else can check and reproduce. That rules out using a neural network as the thing that makes the call, because "the model said so" is not an argument a flight safety review can accept. HATI uses fixed physical formulas with no training data anywhere in the decision path.

4 How to read the numbers

Five kinds of number appear in the figures. This section explains all of them.

4.1 AUC, the score used everywhere

AUC is the single most important number in this document. It answers one question.

Pick one patch of rough ground and one patch of smooth ground at random. Ask the measurement which of the two is rougher. The AUC is the fraction of times it gets that right.

- **1.00** means it is never wrong.
- **0.80** means it is right about four times out of five. Good.
- **0.50** means it is guessing. A coin flip. The measurement carries no information.
- **Below 0.50** means it is reliably wrong in the opposite direction. This is worse than useless, because it actively points you the wrong way.

That last point matters later, so it is worth sitting with. A measurement scoring 0.37 is not “a bit weak”. It ranks smooth ground as rougher than rough ground about 63 % of the time. Feeding it into a hazard model makes the model worse than leaving it out.

4.2 Slope in degrees

How tilted the ground is. 0° is flat. A wheelchair ramp is about 5°. A steep road is around 10°. Landers have a limit, often somewhere near 8° to 15°, past which they risk tipping. Because slope is measured in degrees it means the same thing everywhere, which turns out to be the key property.

4.3 Height error, in metres

When the shadow method estimates how tall a rock is, we can compare its answer against rocks whose true height we already know. The typical size of the mistake is quoted in metres. An error of 0.24 m means the estimates are off by about 24 cm on average.

4.4 False alarm rate

Point the system at ground that is known to be safe and count how often it cries wolf. A false alarm rate of 5 % means it wrongly flags about five patches in every hundred safe ones. Lower is better, but zero is not the goal: a system that never raises an alarm is not detecting anything.

4.5 The z-score

A way of saying how unusual something is compared to its surroundings, counted in “typical spreads”. A z of 0 is perfectly average. A z of 3.22, which is what HATI gives the Athena landing spot, means it is more than three spreads above normal for that area. Unusual, but note carefully that this is unusual *compared to its own neighbourhood*, not unusual in absolute terms. That distinction causes a lot of confusion and is dealt with directly in section 9.

5 Figure 1: which measurements survive a change of site

5.1 What you are looking at

HATI originally measured surface roughness ten different ways. Names like “Slope IQR” and “MDS at 16 m” are just different statistical recipes for “how bumpy is this patch”.

To find out whether those recipes are trustworthy, we ran a controlled test. We took two places on the Moon we already know the answer for: a smooth lava plain in Mare Tranquillitatis, near where Apollo 11 landed, and the rough highland massif of Mons Mouton where Athena came down. A good roughness measurement should score the massif higher than the plain. Every measurement was scored by how often it does that. That score is the AUC from the previous section, and it is what the bars show.

Making the comparison fair took care. Both places were first resampled to the same grid spacing, then blurred so that both carry the same amount of genuine detail. This second step

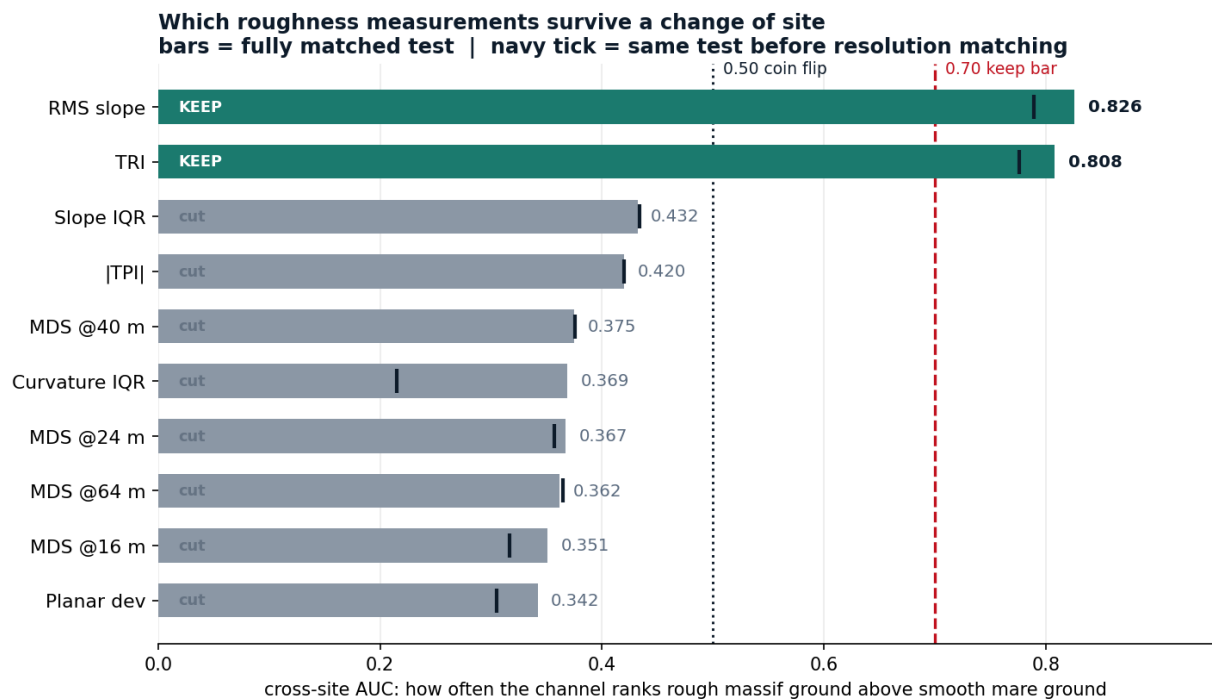


Figure 1: Every roughness measurement HATI originally used, scored on whether it still works when you move it to a different place on the Moon. Green bars cleared the bar and were kept. Grey bars were cut. The navy tick marks where each measurement scored before the test was made fair.

matters more than it sounds. The Apollo 11 map starts at 2 m per pixel and the Mons Mouton map at 4 m, so without that correction the smoother site would look artificially more detailed and the test would be rigged.

5.2 What it means

Only two measurements cleared the bar.

Measurement	AUC	Verdict
RMS slope	0.826	Keep. Right about five times in six.
TRI (ruggedness)	0.808	Keep.
Slope IQR	0.432	Cut. Below a coin flip.
Curvature IQR	0.369	Cut. Below a coin flip.
Planar deviation	0.342	Cut.
TPI	0.420	Cut.
MDS, all four scales	0.35 to 0.38	Cut.

The two that survived have something in common. RMS slope is measured in degrees and TRI in metres. They are *physical* quantities, and a degree is a degree wherever you stand. The eight that failed are all measures of *texture*, statistical descriptions of how varied a patch is, and it turns out those descriptions depend so heavily on the specific place they were tuned on that they do not survive the move.

One result deserves a direct explanation because it is easy to misread. Curvature IQR improved a great deal, climbing from 0.215 to 0.369 once the resolution mismatch was corrected. That is a genuine improvement of 0.153 and it tells us something real about how much resolution matching matters. But it finished at 0.369, which is still below 0.50. It got substantially less wrong without

ever becoming useful. Improvement and usefulness are different things, and only the second one earns a place in the model.

What this tells us

HATI used to have ten roughness measurements. After this test it has two: RMS slope and TRI. The other eight were removed. This makes the system simpler and easier to defend, because every part that remains has evidence behind it.

Honest limit

This is a negative result about our own work, and it was not the outcome we wanted. We report it because a measurement that scores below a coin flip on a controlled test has no business being in a safety system, and because a reviewer would find this in five minutes if we hid it.

6 Figure 2: the slope gate, and where Athena falls

HATI v2.5 absolute layer: physical channels transfer; slope gate is conservative; Athena (gentle) passes -> a sub-resolution / shadow-channel hazard

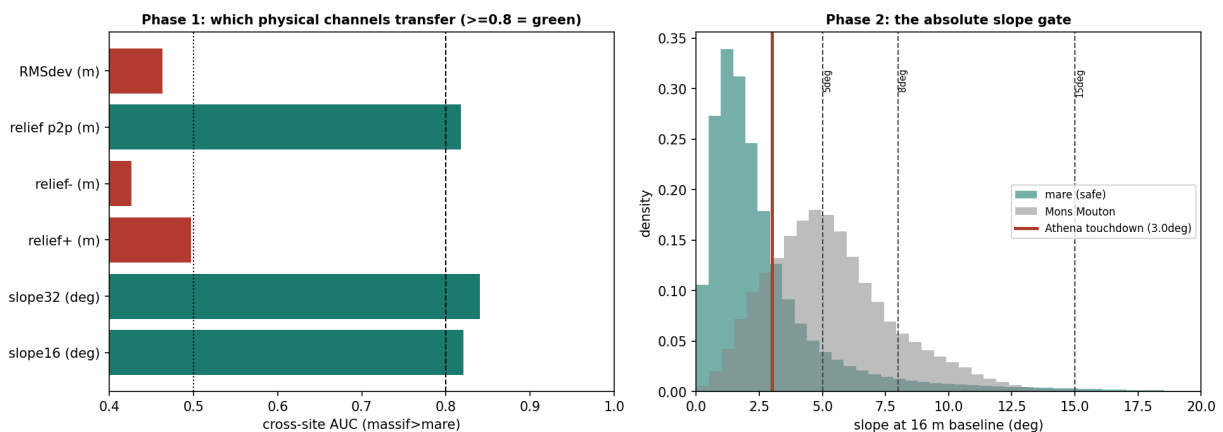


Figure 2: Left: which measurements transfer between sites, the same test as fig. 1 in its earlier form. Right: how steep the ground is at the two sites, drawn on one shared scale, with the lander's limit marked and the Athena landing point shown in red.

6.1 What you are looking at

The right-hand panel is the important one. It shows two histograms, which are just counts of how much ground falls at each steepness. The green shape is the safe lava plain. The grey shape is the rough massif. The dashed vertical lines mark candidate lander limits at 5°, 8° and 15°. The red line marks the exact spot where Athena landed.

6.2 What it means

Set the limit at 8°, which is a reasonable value for a lander of this class. Then:

- 5.26 % of the known-safe plain gets flagged. That is the false alarm rate, and it is an upper bound rather than a true error, because the plain does contain some real crater walls steeper than 8°.
- 14 % of the rough massif gets flagged.
- The Athena landing point measures 3.0° and **passes**. The gate says this ground is fine.

That last line looks at first like a failure. It is not. The gate is doing exactly its job and reporting, correctly, that the ground Athena landed on is not steep. The thing that tipped the lander was a crater below the resolution of the map the gate reads. This is the central finding of

the whole project: the hazard that destroyed the mission was invisible to elevation-based methods in principle, not by accident.

For comparison, the older texture-based approach produced a false alarm rate of about 20 % on the same safe ground. The physical gate is roughly four times more specific.

What this tells us

The slope gate handles big terrain and does it well. It also proves that big terrain was not what went wrong at Athena. That is what sends the whole project towards shadows.

7 Figure 3: two places, one scale

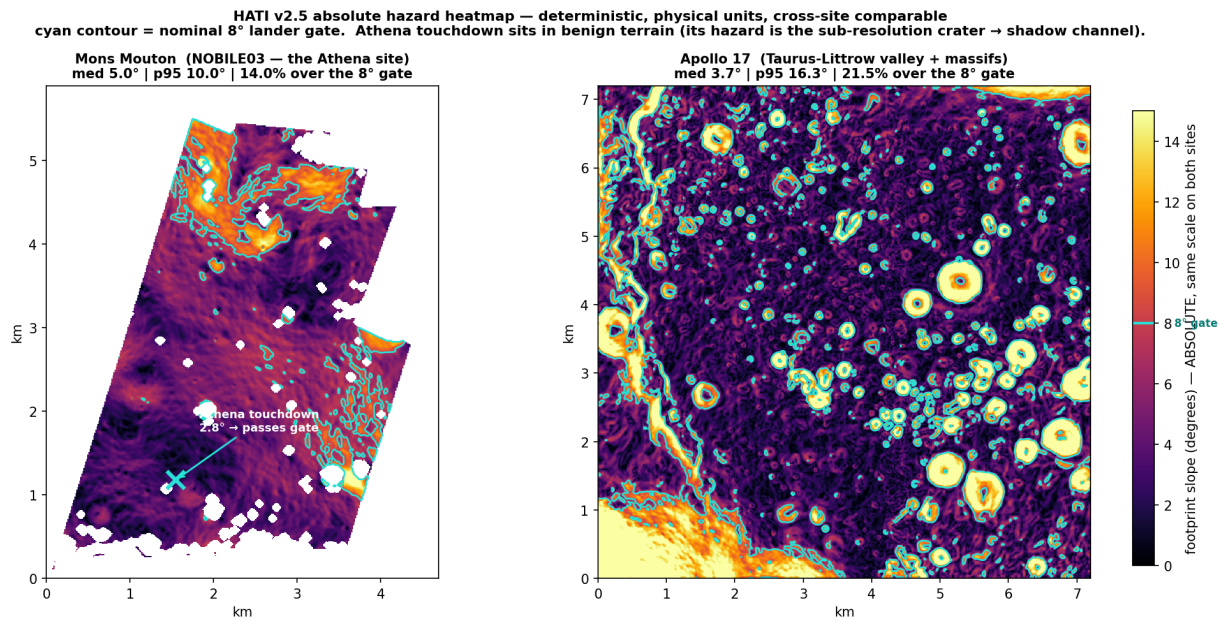


Figure 3: Steepness maps of Mons Mouton, where Athena landed, and the Apollo 17 valley, drawn with the same colours meaning the same thing on both. The cyan outline traces the 8° limit. The cross marks the Athena landing point.

7.1 What you are looking at

Two maps of different places on the Moon. Dark is flat, bright is steep. Both use the same colour scale, so a given brightness means the same steepness on both images.

7.2 What it means

Being able to put those two maps side by side is the point. Because the measurement is in degrees, 7° is 7° in both pictures, and you can compare them directly. In the Apollo 17 panel you can see crater rims and the mountain walls light up, with the valley floor staying dark, which matches what we know that landscape looks like.

The earlier texture-based approach could not support this comparison at all, because it scored each scene relative to itself. In that system a value of 0.7 meant “rough compared to whatever else is in this picture”, which is not a measurement of anything and cannot be carried from one place to another.

8 Figure 4: using the Sun as a probe

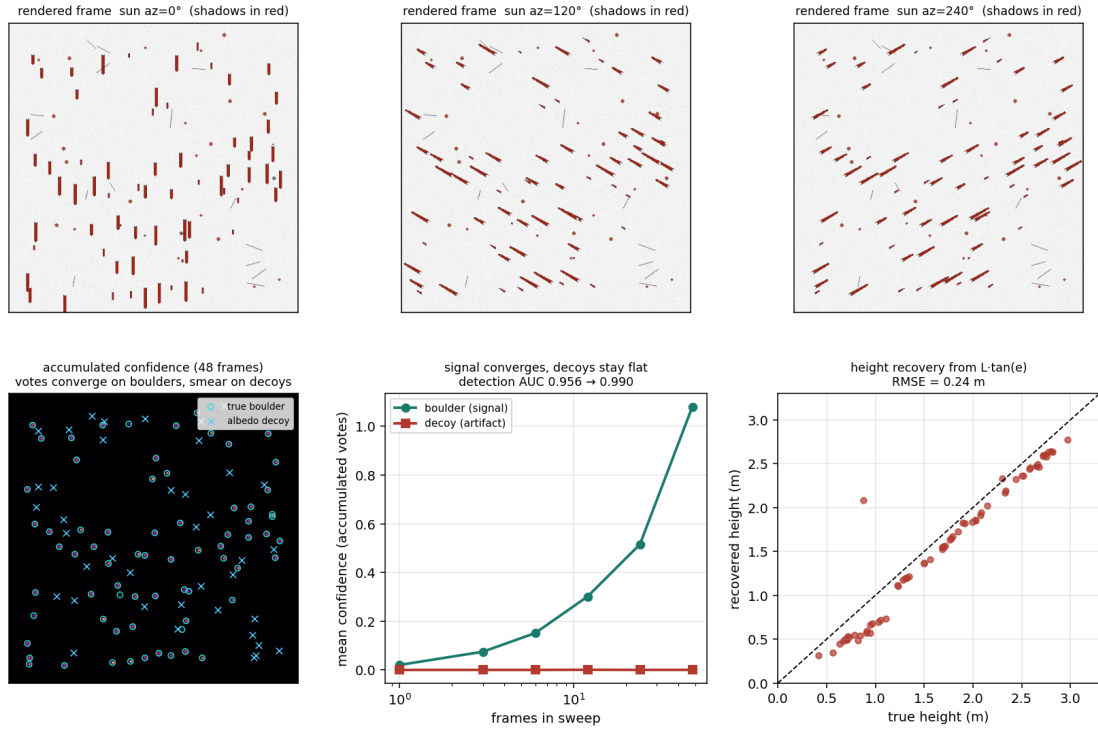


Figure 4: Top row: the same simulated ground lit from three different Sun directions. The red shadows swing around their rocks as the Sun moves. Bottom left: accumulated confidence, bright where the method has decided a real rock stands. Bottom middle: confidence in real rocks grows with more images while confidence in fake ones stays at zero. Bottom right: estimated rock heights against their true heights.

8.1 What you are looking at

This is the method for finding objects too small for the elevation map, tested on simulated ground where we planted the rocks ourselves and therefore know every correct answer.

The scene contains 70 rocks between 0.3 m and 3 m tall. All of them are smaller than the elevation map could resolve. It also contains 40 deliberate traps: dark patches and dark streaks that look exactly like shadows in any single photograph but are simply dark rock.

8.2 What it means

The trick is to use the Sun’s movement. Over a lunar day the Sun circles the horizon, so the same ground gets photographed under many different lighting directions. A real rock’s shadow swings around it like the shadow of a sundial gnomon, always anchored at the rock’s base. A dark patch of ground does not move at all, because it is dark for chemical reasons rather than geometric ones.

So the method drops a vote at the base of every shadow it sees, in every image. Real rocks collect votes at the same point every time and build up a sharp bright spot. Fake ones scatter their votes in a ring that never adds up to anything. You can see exactly this in the bottom middle panel: the green line, which is real rocks, climbs steadily with more images, while the red line, which is the traps, stays flat on zero.

Once a rock is confirmed, eq. (1) converts its shadow length into a height. The bottom right panel compares those estimates against the truth. The typical error is 0.24 m, for rocks between 0.3 m and 3 m tall.

Honest limit

Everything in this figure is simulated. The method works on ground we invented, where the terrain is flat, the traps come in only two varieties, and the images line up perfectly. Real images have none of those luxuries. Until this runs on real photographs, it is a promising estimator and nothing more. The archive check has been done, so we know the images exist: 696 photographs of the Athena site, 525 of them with the Sun up, covering every direction of the compass.

9 What HATI can and cannot say

Statement	Status	Evidence
The Athena spot is unusual compared to its surroundings	Established	$z = 3.22$, real data
Steepness and ruggedness carry between sites	Established	AUC 0.81 to 0.83
The slope gate is specific on safe ground	Established	5.26 % false alarm
Athena's ground was not steep, so the hazard was small-scale	Established	3.0° measured
Texture measurements do not carry between sites	Established, negative	AUC 0.34 to 0.43
Shadows can separate real rocks from dark patches	Simulated only	Traps score zero
Shadows can measure rock height	Simulated only	0.24 m typical error
The same works on real photographs	Not yet done	Images confirmed to exist
A calibrated probability of losing a mission	Not attempted	See below

Two limits deserve saying in full.

HATI does not produce a probability that a landing will fail. It produces a hazard ranking. Turning a ranking into a real probability needs many recorded landing outcomes to calibrate against, and humanity has only a handful of instrumented lunar landings. Anyone quoting a percentage chance of failure from this system is overstating what the data supports.

The $z = 3.22$ figure for the Athena site says that spot is unusual *for that neighbourhood*. On an absolute scale, measured in degrees and metres, the same ground is unremarkable. Both statements are true and they are not in conflict, because they measure different things. The absolute measurement is the one that transfers between sites, and it is the one to trust for site selection.

10 Poseidon: the same idea, at sea

The interesting thing about the shadow method is that almost nothing in it is about the Moon. Strip away the setting and what remains is a general recipe:

Find objects too small or too faint to see directly, by collecting evidence across many observations taken under different geometry, where physics predicts how a real object's signature must change between observations. Reject anything that fails to change the way physics says it should.

Project Poseidon applies that recipe to maritime surveillance. The problem there is dark vessels: ships that switch off their transponders, which is exactly when they most want watching. Illegal fishing boats do it. Sanctioned tankers do it. So do vessels loitering above undersea cables and pipelines, which since the Nord Stream attacks has become a serious concern in the Baltic and North Sea.

10.1 The physical bridge

The transfer is closer than it looks, and it is not a loose analogy.

On the Moon, the length of a shadow encodes the height of the rock that cast it, by eq. (1). At sea, radar satellites have a comparable quirk. In a synthetic aperture radar image, a moving target is displaced sideways in the picture by an amount proportional to how fast it is moving towards or away from the satellite:

$$\Delta x \approx \frac{R}{V} v_r \tag{2}$$

where R is the range to the target, V the satellite’s speed, and v_r the ship’s speed along the line of sight. A moving ship therefore appears offset from its own wake, and the size of that offset tells you its velocity.

The shapes of eq. (1) and eq. (2) are the same. One physical property of an object, recovered from a geometric displacement, using nothing but known geometry. The auditability that makes HATI defensible carries across unchanged.

HATI, on the Moon	Poseidon, at sea
The Sun moves through many angles	The satellite revisits at many angles
Shadow length gives rock height	Wake offset gives ship velocity
Dark rock that never moves is rejected	Sea clutter that ignores physics is rejected
Votes converge at a rock’s base	Detections stay consistent across passes
Slope gate handles big terrain	Standard clutter filtering handles easy cases
Cross-checked against Diviner rock counts	Cross-checked against transponder records

10.2 Why the sea is the easier place to prove it

There is a real strategic advantage here. HATI has essentially one instrumented landing to check itself against. Poseidon has transponder data, which is public, continuous, and covers thousands of ships every week.

That flips the usual relationship. Match a radar detection against the transponder records and you can measure precisely how often the method is right. And the detections that have *no* matching transponder record are not errors to be explained away. They are the product, because a ship the radar sees and the transponder does not report is exactly the dark vessel a coast guard wants to know about.

The practical route is short. The European Sentinel-1 satellites provide radar images of the sea free of charge, day, night and through cloud (Torres et al. 2012). Ship detection in radar imagery is a mature field with established baselines to measure against (Crisp 2004). A first test needs one radar scene over a stretch of the Baltic, the matching transponder records for the same hours, and a comparison of the two.

Honest limit

Some things do not carry over, and pretending otherwise would be dishonest. Radar physics is not shadow physics, so the front end of the detector has to be rebuilt. The sea surface moves, unlike lunar terrain. Radar images carry speckle, a grainy interference pattern with statistics quite unlike ordinary image noise. What transfers is the method and the discipline, not the code.

11 Glossary

AUC

How often a measurement correctly picks the rougher of two random patches. 0.5 is guessing, below 0.5 is reliably wrong.

DEM

Digital Elevation Model. A map storing the height of the ground at each pixel.

Effective resolution

The size of the smallest feature a map can genuinely show, usually several times coarser than its pixel spacing suggests.

False alarm rate

How often a system flags ground that is actually safe.

NAC

Narrow Angle Camera, the high-resolution camera on NASA's Lunar Reconnaissance Orbiter (Robinson et al. 2010).

Orthophoto

A photograph corrected so that distances in the image match distances on the ground.

RMS slope

A measure of average steepness across a patch, in degrees.

SAR

Synthetic Aperture Radar. A radar that builds a detailed image using the satellite's own motion, and works through darkness and cloud.

Solar sweep

A set of photographs of the same place taken under many different Sun directions.

TRI

Terrain Ruggedness Index. The average height difference between a point and its immediate neighbours, in metres (Riley et al. 1999).

z-score

How unusual a value is compared to its surroundings, counted in typical spreads.

12 Sources and further reading

The roughness measurements HATI started from come from the planetary science literature, chiefly the work on lunar and martian surface roughness by Kreslavsky and Head (2000), Rosenberg et al. (2011) and Kreslavsky et al. (2013), with later refinements in Cai and Fa (2020). The ruggedness index is older and comes from terrestrial geomorphology (Riley et al. 1999).

The imagery and elevation data are from the Lunar Reconnaissance Orbiter Camera (Robinson et al. 2010), with the stereo elevation models built by the method of Henriksen et al. (2017). Where surface brightness needs a physical model, the standard reference is Bruce Hapke (1984), with lunar parameters mapped by Sato et al. (2014). The limit on what any sampled grid can represent is the classic sampling result of Shannon (1949).

For the maritime work, the Sentinel-1 radar mission is described by Torres et al. (2012) and the state of ship detection in radar imagery by Crisp (2004).

A note on the references

The two maritime references were recalled from memory rather than looked up, and are marked as unverified in the bibliography file. Confirm the author lists, volumes and page numbers before this document is submitted anywhere. The lunar references have been carried through earlier HATI papers, but a fresh check costs little.

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